Telecommunication Systems
ICE 3109

Lecture 9
GSM Architecture

Introduction

- **GSM = Global System for Mobile communications**
- (**GSM**: originally from *Groupe Spécial Mobile*)
- Driving Factors:
  - Incompatibility of the European analog cellular systems
  - Reaching of capacity limits
  - Costs of the equipment

- 1982, Conference of European Post and Telecommunications formed Group Speciale Mobile (GSM)
- 1987, 15 operators from 13 countries signed Memorandum of Understanding (MoU)
- 1991, Finland's operator Radiolinia launched first GSM network in July 1991
- 1992, Massive deployment of GSM started
- By 2000 GSM became the most popular 2G technology worldwide
- GSM standard still evolving and enriched with new features and services

GSM Frequency Resources

GSM900 :
up: 890~915MHz
down: 935~960MHz
duplex interval: 45MHz
bandwidth: 25MHz，
frequency interval: 200KHz

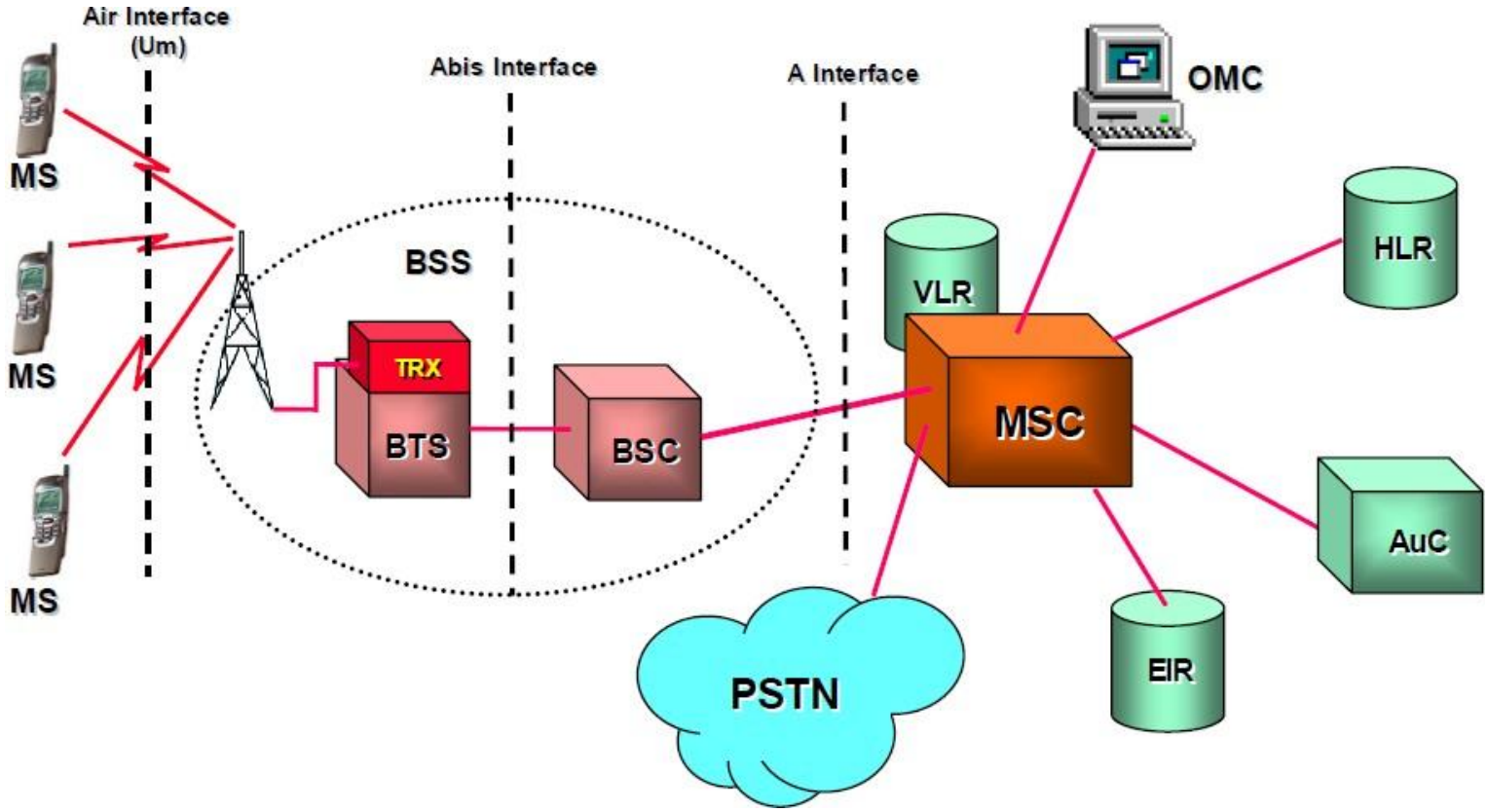
EGSM900 :
up: 880~890MHz
down: 925~935MHz
duplex interval: 45MHz
bandwidth: 10MHz，
frequency interval: 200KHz

GSM1800 :
up: 1710-1785MHz
down: 1805-1880MHz
duplex interval: 95MHz，
working bandwidth: 75MHz，
frequency interval: 200KHz

GSM1900MHz:
up:1850~1910MHz
down:1930~1990MHz
duplex interval: 80MHz，
working bandwidth: 60MHz，
frequency interval: 200KHz

GSM Architecture

- Overview of GSM Architecture



A GSM network is made up of three subsystems:

- The Mobile Station (MS)
- The Base Station Sub-system (BSS) – comprising a BSC and several BTSs
- The Network and Switching Sub-system (NSS) – comprising an MSC and associated registers

Several interfaces are defined between different parts of the system:

- 'A' interface between MSC and BSC
- 'Abis' interface between BSC and BTS
- 'Um' air interface between the BTS (antenna) and the MS

Abbreviations:

MSC – Mobile Switching Centre
BSC – Base Station Controller
BTS – Base Transceiver Station
TRX – Transceiver
MS – Mobile Station
BSS – Base Station Sub-system
HLR – Home Location Register
VLR – Visitor Location Register
AuC – Authentication Centre
EIR – Equipment Identity Register
OMC – Operations and Maintenance Centre
PSTN – Public Switched Telephone Network

The Mobile Station

- The mobile station consists of:
  - mobile equipment (ME)
  - subscriber identity module (SIM)

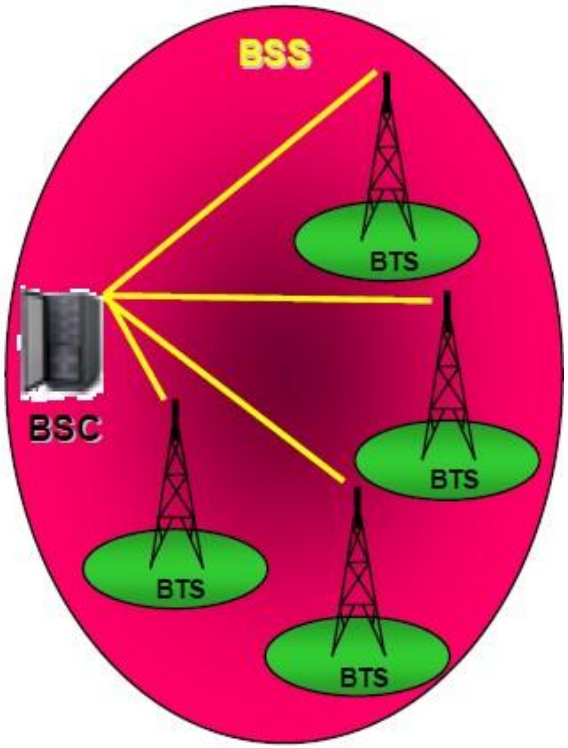


- The SIM stores permanent and temporary data about the mobile, the subscriber and the network, including:
  - The International Mobile Subscribers Identity (IMSI)
  - MS ISDN number of subscriber
  - Authentication key (K_i) and algorithms for authentication check



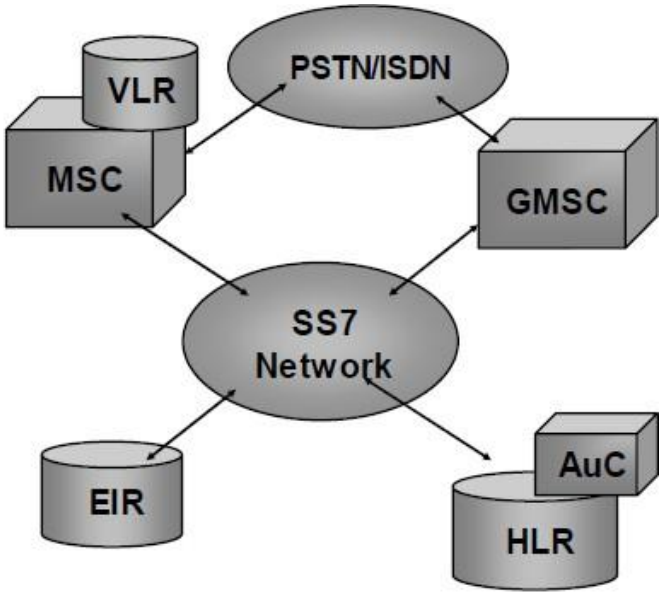
Base Station Sub-system (BSS)

- The BSS comprises:
  - Base Transceiver Station (BTS)
  - One or more Base Station Controllers (BSC)
- The purpose of the BTS is to:
  - provide radio access to the mobile stations
  - manage the radio access aspects of the system
- BTS contains:
  - Radio Transmitter/Receiver (TRX)
  - Signal processing and control equipment
  - Antennas and feeder cables
- The BSC:
  - allocates a channel for the duration of a call
  - maintains the call:
    - monitoring quality
    - controlling the power transmitted by the BTS or MS
    - generating a handover to another cell when required
- Siting of the BTS is crucial to the provision of acceptable radio coverage



Network Switching System (NSS)

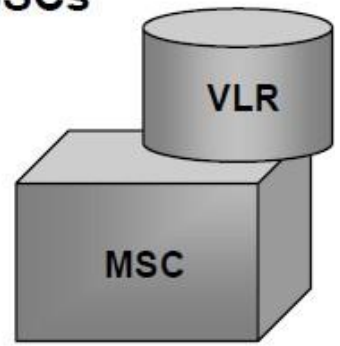
- Key elements of the NSS:
  - Mobile Switching Centre (MSC) with:
    - Visitor Location Register (VLR)
  - Home Location Register (HLR) with:
    - Authentication Centre (AuC)
  - Equipment Identity Register (EIR)
  - Gateway MSC (GMSC)



- These elements are interconnected by means of an SS7 network

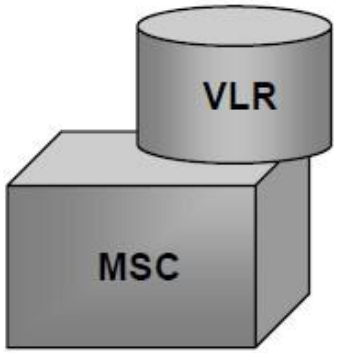
Mobile Switching Centre

- **Functions of the MSC:**
  - **Switching calls, controlling calls and logging calls**
  - **Interface with PSTN, ISDN, PSPDN**
  - **Mobility management over the radio network and other networks**
  - **Radio Resource management - handovers between BSCs**
  - **Billing Information**



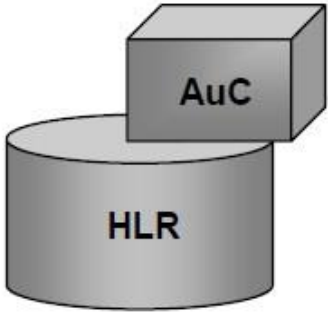
Visitor Location Register (VLR)

- Each MSC has a VLR
- VLR stores data temporarily for mobiles served by the MSC
- Information stored includes:
  - IMSI
  - Mobile Station ISDN Number
  - Mobile Station Roaming Number
  - Temporary Mobile Station Identity
  - Local Mobile Station Identity
  - The location area where the mobile station has been registered
  - Supplementary service parameters



Home Location Register (HLR)

- Stores details of all subscribers in the network , such as:
  - Subscription information
  - Location information: mobile station roaming number, VLR, MSC
  - International Mobile Subscriber Identity (IMSI)
  - MS ISDN number
  - Tele-service and bearer service subscription information
  - Service restrictions
  - Supplementary services



- Together with the AuC, the HLR checks the validity and service profile of subscribers

Equipment Identity Register (EIR)

- ✓ Validates mobile equipment
- ✓ Invalid IMEI if handset is stolen

The Authentication Centre (AUC)

- ✓ used for the authentication a mobile subscriber that wants to be connected to the network.

Gateway Mobile Switching Centre (GMSC)

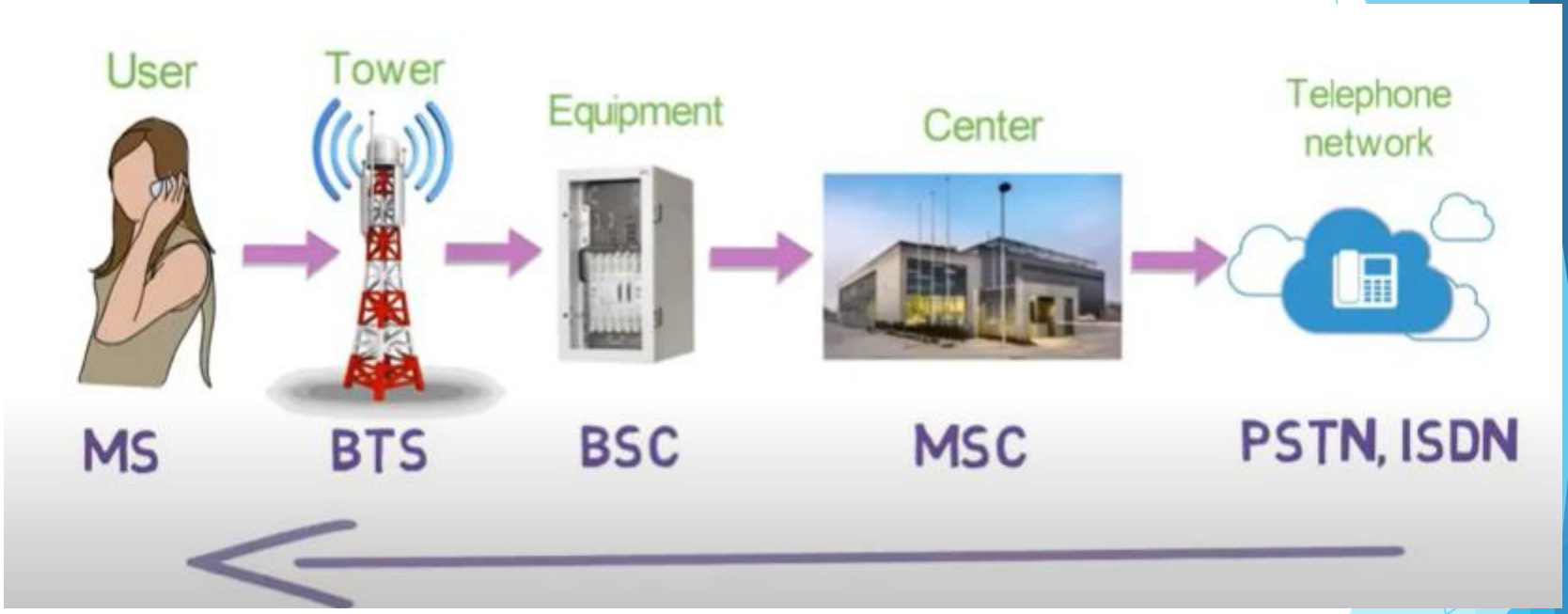
- A Gateway Mobile Switching Centre (GMSC) is a device which routes traffic entering a mobile network to the correct destination
- The GMSC accesses the network's HLR to find the location of the required mobile subscriber
- A particular MSC can be assigned to act as a GMSC
- The operator may decide to assign more than one GMSC



GSM protocol Architechure

- GSM protocols are basically divided into three layers:
  - Layer 1: Physical layer
    - Enables physical transmission (TDMA, FDMA, etc.)
    - ± Assessment of channel quality
    - ± Except on the air interface (GSM Rec. 04.04), PCM 30 or ISDN
    - links are used (GSM Rec. 08.54 on Abis interface and 08.04 on A to F interfaces).
  - Layer 2: Data link layer
    - Multiplexing of one or more layer 2 connections
    - on control/signaling channels
    - Error detection (based on HDLC)
    - Flow control
    - Transmission quality assurance
    - Routing
  - Layer 3: Network layer
    - Connection management (air interface)
    - Management of location data
    - Subscriber identification
    - Management of added services (SMS, call forwarding, conference calls, etc.)

Call Setup in GSM



GSM Architecture

- Overview of GSM Architecture

